C 20210 **(Pages : 3)** **Name.....**

Reg. No.....

SIXTH SEMESTER (CUCBCSS–UG) DEGREE EXAMINATION, MARCH 2022

Mathematics

MAT 6B 12—NUMBER THEORY AND LINEAR ALGEBRA

(2014 to 2018 Admissions)

Time : Three Hours Maximum : 120 Marks

Section A

Answer **all** *questions.*
Each question carries 1 *mark.*

1. State general version of the Division Algorithm.
2. Find lcm (306, 657).
3. State the Fundamental Theorem of Arithmetic.
4. Explain the method of Sieve of Eratosthenes.
5. Define Pseudoprime. Give an example of a Pseudoprime number.
6. Find $\sigma(12)$.
7. Define Euler's Phi Function.
8. Prove that the set $X = \{(x, 0) : x \in \mathbb{R}\}$ is a subspace of the vector space $\mathbb{R}^2$.
9. Define basis of a vector space.
10. Check whether the map $f : \mathbb{R}^3 \to \mathbb{R}^3$ defined by $f(x, y, z) = (x + y, z, 0)$ is linear. Justify your claim.
11. Define the rank and nullity of the linear map.
12. For $(x, y) \in \mathbb{R}^2$, let $S = \{(x, y)\}$. Find $\langle S \rangle$.

(12 × 1 = 12 marks)

Section B

Answer any **ten** *questions.*
Each question carries 4 *marks.*

13. Show that square of any odd integer is of the form $8k + 1$.
14. If $a \mid c$ and $b \mid c$, with $gcd(a, b) = 1$, prove that $ab \mid c$.
15. Use the Euclidean Algorithm to obtain integers x and y satisfying $gcd(24, 138) = 24x + 138y$.
16. Determine all solutions in the positive integers of the Diophantine equation $18x + 5y = 48$.

Turn over

17. Prove that the number $\sqrt{2}$ is irrational.

18. For arbitrary integers a and b, prove that $a \equiv b(\text{mod}\, n)$ if and only if a and b leave the same non-negative remainder when divided by n.

19. Use the binary exponential algorithm to compute $5^{110} \pmod{131}$.

20. Solve the linear congruence $18x \equiv \pmod{42}$.

21. If p and q are distinct primes with $a^p \equiv a(\text{mod}\, q)$ and $a^q \equiv a(\text{mod}\, p)$, then prove that $a^{pq} \equiv a(\text{mod}\, pq)$.

22. Prove that intersection of two subspaces of vector space V over a field is again a subspace.

23. Let A = Span {(1, 2, 0, 1), (– 1, 1, 1, 1)} and B = Span {(0, 0, 1, 1), (2, 2, 2, 2)} be two subspaces of $\mathbb{R}^4$. Determine $\text{A} \cap \text{B}$ and compute its dimension.

24. If V is a vector space over the set $\mathbb{C}$ of complex numbers of dimension n, prove that V can be regarded as a vector space over $\mathbb{R}$ of dimension $2n$.

25. Let $f : \text{V} \to \text{W}$ be linear. Prove that if Y is subspace of W then $f^{\leftarrow}(\text{Y})$ is a subspace of V.

26. Let V be a vector space of dimension $n \geq 1$ over a field F. Prove that V is isomorphic to the vector space F^n.

(10 × 4 = 40 marks)

Section C

Answer any **six** *questions.*
Each question carries 7 *marks.*

27. Given integers a and b, not both of which are zero, prove that there exists integers x and y such that $gcd\ (a, b) = ax + by$.

28. Prove that the linear Diophantine equation $ax + by = c$ has an integer solution if and only if $d \mid c$ where $d = \gcd(a, b)$.

29. Prove that there are infinite number of primes.

30. Using Chinese Remainder Theorem, solve the system of congruences $x \equiv 5(\text{mod}\, 11), x \equiv 14(\text{mod}\, 29), x \equiv 15(\text{mod}\, 31)$.

31. If p is a prime, then prove that $(p-1)! \equiv -1(\text{mod}\, p)$.

32. Prove that a non-empty subset S of a vector space V is a basis of V if and only if every element of V can be expressed in a unique way as a linear combination of elements of S.

33. Let $f, g, h : \mathbb{R} \to \mathbb{R}$ be the mappings given by $f(x) = \cos^2 x, g(x) = \sin^2 x, h(x) = \cos 2x$. Consider the subspace of $\text{Diff}(\mathbb{R}, \mathbb{R})$ given by W = Span {f, g, h}. Find a basis for W.

34. Show that the linear mapping $f : \mathbb{R}^3 \to \mathbb{R}^3$ given by $f(x, y, z) = (x + y + z, 2x - y - z, x + 2y - z)$ is both surjective and injective.

35. State and prove Dimension Theorem.

(6 × 7 = 42 marks)

Section D

Answer any **two** *questions.*
Each question carries 13 *marks.*

36. (a) If $a = qb + r$, prove that $\gcd(a, b) = \gcd(b, r)$. (4 marks)

(b) If a cock is worth 5 coins, a hen 3 coins, and three chicks together 1 coin, how many cocks, hens, and chicks, totalling 100, can be bought for 100 coins.

(9 marks)

37. Prove that the lienar congruence $ax \equiv b \pmod{n}$ has a solution if and only if $d \mid b$, where $d = \gcd(a, n)$. Moreover if $d \mid b$, prove that it has d mutually incongruent solutions modulo n.

38. Prove that every linearly independent subset I of a finite dimensional vector space V can be extended to form a basis.

[2 × 13 = 26 marks]